

SCyPS

The center's newsletter: research, people, and what comes next.



The center's work runs from the power grid and the fiber network to the road and the reactor, with people and AI sharing control.

FROM THE DIRECTOR

Why a newsletter, and why now

Welcome to the first issue. The Center for Smart Cyber-Physical Systems was founded in October 2019 by five faculty who wanted to work on one problem together: making the systems people depend on, from the grid to the road to the hospital, secure and resilient as software and AI take over more of how they run. Seven years on, the center has 19 faculty across four colleges and 215 peer-reviewed papers, and every paper and award is listed on our website with a link to its source, so anyone can check the record.

This fall is a turning point. In October, the National Science Foundation's \$2 million SUMMIT award begins, giving us a federated testbed that links UMass Lowell, NYU, and West Virginia University in one instrument for studying attacks on the grid and recovery from them. This fall we are applying to the university for full research center status, and we are broadening what we do to put people in the loop: operators, drivers, patients, and the trust between them and the automation they work with.

Each month this newsletter will carry one student story, one faculty story, and one project, along with what was published and funded. If you are a student looking for a group, a company looking for a testbed, or a colleague looking for a collaborator, the last page tells you how to reach us. Thank you for reading.

Vinod M. Vokkarane, Director

In this issue

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By the numbers

19

faculty, four
colleges

215

papers since 2019

87

in journals

16

doctoral students

Eight thrusts, one loop

Cyber-physical systems sense the physical world, communicate what they sense, decide, and act. The center's research covers every link in that loop, and the eight thrusts below are how its 19 faculty organize the work. Each thrust has a lead, a page on the center site, and a record of papers and awards that the site keeps current every week.

Smart grid cybersecurity and resilience

Vokkarane, Arias, Tseng, Lin

AI and agentic systems for cyber-physical control

Luo, Cao, Vokkarane, Son

Next-generation optical and 6G transport

Vokkarane, Chigan

Fault-tolerant distributed and edge computing

Tseng, Luo

Hardware security and trusted devices

Arias, Akyurtlu, Ranasingha

High performance computing and data integrity

Son, Luo

Connected transportation, health, and infrastructure

Xie, Luo, Cao, Yu, Inalpolat, Robinette, Niezrecki

Nuclear energy and security

Aghara, Niezrecki

Active and new awards (18)

- **SUMMIT**
National Science Foundation, Major Research Instrumentation Track 2, \$2.0M
- **ARPO-Sensor Fusion**
Massachusetts Technology Collaborative, Applied AI Models program, \$625K
- **BOND-AI**
NextFlex, \$1.0M
- **ARPO: Autonomous Robotic Planning and Optimization**
U.S. Army, \$225K
- **CyberCARED**
U.S. Department of Energy, \$3.5M
- **Massachusetts Advanced Nuclear and Fusion Energy Roadmaps**
Commonwealth of Massachusetts, Healey-Driscoll Administration
- **Intercontinental Nuclear Institute (INI)**
International Atomic Energy Agency
- **Advanced Technologies and Data Analytics for Safe, Smart, and Efficient Transportation (ASSET)**
Massachusetts Department of Transportation, \$604K
- **A Railroad Trespassing Detection and Dynamic Warning System Based on Artificial Intelligence and Edge Computing**
Massachusetts Department of Transportation, \$750K
- **Maine Advanced Signal Control and Connected Vehicle System for Safe, Efficient and Equitable Rural Transportation (MAST)**
U.S. Department of Transportation, ATCMTD, through MaineDOT, \$150K
- **Planning Federated AI-Ready Cyberinfrastructure for Advanced Microscopy and Imaging**
National Science Foundation, OAC Category III, \$499K
- **Towards Fault-tolerant Edge Computing for Cyber-Physical Systems**
National Science Foundation, CNS CAREER, \$342K
- **Improving Data Integrity for HPC Datasets using Sparsity Profile**
National Science Foundation, OAC Core, \$600K
- **Open-Source Research: Friendly Fedora and Podman**
Red Hat Inc., \$200K+
- **Resilient Smart Grids**
Navia Energy Inc.
- **BioSPACE**
National Science Foundation, PFI-RP, \$1.0M
- **Unsupervised Deep PCCT Reconstruction for Human Extremity Imaging**
National Institutes of Health, \$2.30M
- **AI-based Cardiac CT**
National Institutes of Health, \$2.4M

Full records: smartcyberphysical.org. Publications, awards, and Scholar figures on the site are refreshed automatically from Crossref, NSF, and Google Scholar.

SUMMIT: a testbed for attacking, defending, and recovering the grid

A \$2.0 million NSF Major Research Instrumentation award builds a three-university instrument where real control, network, and security hardware closes the loop with a real-time simulation of the Northeast transmission grid.



SUMMIT's architecture: the UMass Lowell main site with its real-time digital simulators and control, network, and instrumentation equipment; a wide-area software-defined network over the Internet; and the federation sites at NYU Tandon and West Virginia University.

Grid cybersecurity research has a practical problem: you cannot attack a real grid to see what happens, and a pure software simulation leaves out the relays, controllers, and network devices where attacks actually land. SUMMIT is built to close that gap. Funded by NSF's Major Research Instrumentation program (award #2511635) at \$2.0 million, with a UMass Lowell share of \$1.56 million and subawards to NYU and West Virginia University, it runs from October 2026 through September 2029.

What it is

At the core are RTDS NovaCor real-time digital simulators, which run grid models at time steps small enough to drive real relays, controllers, and network devices in closed loop. Every power component, from generation and storage to inverters, lives inside the simulation. Around the simulators at the UMass Lowell main site sit a signal generator for disturbance injection, a network emulator for latency, loss, and attack scenarios, optical, RF, and FPGA equipment for the transport and edge layers, and GPS-synchronized control and protection equipment. Wide-area software-defined network links connect the three sites over the public Internet.

Four paradigms

SUMMIT supports four ways of working: distributed hardware-in-the-loop simulation, where real control, network, and security hardware closes the loop with grid models running in real time; Internet-in-the-loop simulation, where the three sites exchange live signals so the network itself is part of every experiment; distributed digital twins, scoped in Phase 1 to the RTDS simulators and the center's OPAL-RT simulator; and the central objective, a federated platform that delivers hardware-in-the-loop Simulation-as-a-Service to collaborators who never touch the hardware.

Phase 1, and who it is for

Phase 1 models the Northeast's backbone transmission grid at proof-of-concept scale, with every capability needed to scale to regional and national models in Phase 2. The team is PI Vinod Vokkarane; Co-PIs Orlando Arias and Lewis Tseng (UMass Lowell), Yuzhang Lin (NYU), and Anurag Srivastava (WVU); and senior personnel Yan Luo, Seung Woo Son, and Christopher Niezrecki. A postdoctoral researcher will lead federation development; the position is open. Collaborators can propose experiments as the instrument opens in 2026 to 2027.

Sukesh Aghara: the Commonwealth's nuclear roadmap and the center's energy thrust

A center co-founder who now advises the Chancellor on nuclear energy strategy, leads the Massachusetts Advanced Nuclear and Fusion Energy Roadmap, and directs the laboratory where nuclear security meets cyber-physical systems.

When the center was founded in 2019, Sukesh Aghara was one of its five co-founders and its founding co-director for energy. He still leads the nuclear energy and security thrust. What has changed is the scale of the questions he is asked to answer. In October 2025 he was asked to lead the Massachusetts Advanced Nuclear and Fusion Energy Roadmap, the Commonwealth's plan for how advanced reactors and fusion fit its energy future. In 2026 UMass Lowell named him Senior Advisor to the Chancellor on Nuclear Energy Strategies, after five years as the Francis College of Engineering's Associate Dean for Research and eight as director of the Nuclear Engineering Program.



Sukesh Aghara, Professor of Chemical and Nuclear Engineering and Senior Advisor to the Chancellor on Nuclear Energy Strategies.

The laboratory

Aghara directs the Integrated Nuclear Security and Safeguards Laboratory, INSSL, which sits alongside the UMass Lowell research reactor. The lab has radiation detectors and cyber-physical testbeds, and its work on the cybersecurity of nuclear facilities, safeguards verification, and advanced reactor modeling is supported by federal agencies including the National Nuclear Security Administration. He also co-directs the Intercontinental Nuclear Institute, a partnership with the International Atomic Energy Agency that trains the next generation of nuclear security professionals. Both are where the center's cyber-physical work meets the nuclear world: a reactor's control and protection systems are cyber-physical systems whose failure has physical consequences.

The scholarship

He is the editor of The Oxford Handbook of Nuclear Security, published by Oxford University Press, with chapters of his own on cybersecurity for nuclear facilities and on security for next-generation reactors. His recent work ranges from stochastic methods for safeguards inspection to a June 2026 paper in Energy Reports, with Korinek, Annadevula, and Skoda, on phasing clean alternatives into a university campus's central heating. He chaired the Nuclear Engineering Department Heads Organization in 2024 to 2025.

This fall

On September 30, UMass Lowell hosts the New England Governors' Nuclear Energy Summit at the Rist Institute. For the center, it is where the SUMMIT testbed meets the nuclear community: a plant's connection to the grid can be tested in the instrument before the plant is built.

At a glance

- Professor, Chemical and Nuclear Engineering
- Senior Advisor to the Chancellor on Nuclear Energy Strategies (2026)
- Lead, Massachusetts Advanced Nuclear and Fusion Energy Roadmap
- Director, Integrated Nuclear Security and Safeguards Laboratory
- Co-Director, Intercontinental Nuclear Institute (with the IAEA)
- Editor, The Oxford Handbook of Nuclear Security
- Associate Dean for Research, 2021 to 2026

Building the networks behind AI

Three journal articles in six months on how fiber-optic networks can keep up with AI, and an open-source platform the field can build on.



Arash Rezaee

Ph.D. Candidate, joined 2022



Ryan McCann

Ph.D. Candidate, joined 2024

Advised by Vinod M. Vokkarane, Advanced Communication Networks Laboratory

Every AI application, cloud service, and connected device depends on something most people never see: the communication networks carrying enormous amounts of data behind the scenes. Two doctoral researchers in the center's Advanced Communication Networks Laboratory, Arash Rezaee and Ryan McCann, have spent the past year working on how those networks keep up.

Their work has produced three articles accepted in the last six months by the Journal of Optical Communications and Networking, published jointly by IEEE and Optica. Arash is first author on all three. Ryan co-led the work on FUSION, the group's open-source optical networking platform, and contributed to the other two studies. Collaborators at the University of Virginia and in Spain joined the work. Both students are advised by Vinod Vokkarane, the center's director.

The question

Future AI systems will need more than powerful processors and large data centers. They will need networks that move huge volumes of information quickly and reliably between users, edge devices, cloud platforms, and computing facilities. There are two main ways to add capacity to a fiber network: use a wider range of the optical spectrum by transmitting across several wavelength bands, or add spatial capacity by sending data through multiple cores inside one fiber or through parallel fiber pairs.

Both work. Combining them is hard. Signals interfere with one another, performance varies across the spectrum, and some of the added capacity turns out to be unusable. The three papers take on that problem from three directions: long-term planning, real-time operation, and reproducible research.

Planning: capacity on paper is not capacity in practice

The first article develops a planning framework that evaluates optical bands and spatial channels together rather than as separate upgrades. It accounts for signal noise, nonlinear effects, power changes, equipment limits, and interference between neighboring fiber cores, and parts of it were validated on a commercial multicore fiber. Its central finding is that adding bands or cores does not always add usable capacity; the best choice depends on the spectrum in use, the structure of the fiber, and the physical conditions in the network. The article is an invited extension of the team's paper at ECOC 2025.

Operation: smarter decisions while the network runs

The second article turns to the decisions a network makes as traffic arrives. A quality-of-transmission-aware controller chooses the route, band, spatial lane, frequency channel, and modulation format for each connection, grooms new traffic into unused capacity on existing connections, and slices large requests across channels when no single channel can carry them. Comparing multi-core fibers with parallel fiber pairs, the paper shows the conditions under which each performs best rather than declaring a winner. It extends the team's paper at ONDM 2025.

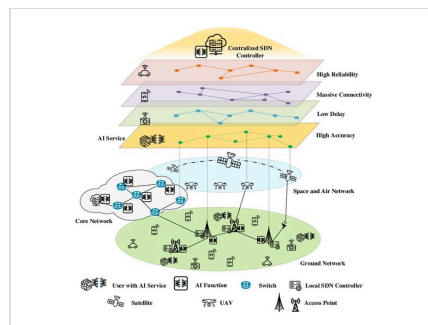
Research: a shared platform, FUSION

The third article introduces FUSION, the Flexible Unified Simulator for Intelligent Optical Networking, which Ryan co-founded and leads as its principal developer. Optical

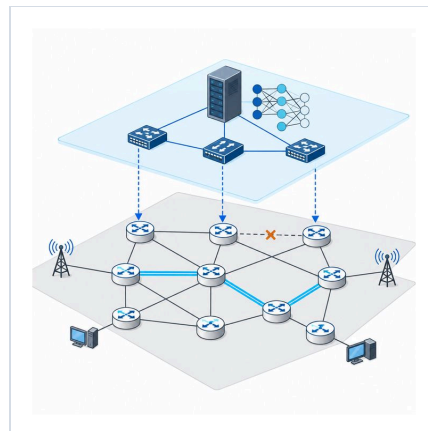
networking studies are hard to compare because groups use different software, assumptions, and network configurations. FUSION gives researchers one modular environment covering conventional, multi-band, and space-division multiplexed networks, with physical-layer models, control algorithms, grooming, survivability, and AI-based decision making, plus automated testing and deterministic replay so experiments can be checked and repeated. It was the platform behind the other two papers, and it is released as open source for others to extend. The article appears in JOCN's feature issue on benchmarking in optical networks.

Why it matters to the center

Smart grids, autonomous platforms, connected health care, and intelligent transportation all depend on networks that move data reliably and respond when conditions change. Arash and Ryan's work connects fiber physics to network control and to software the field can share. It is the communications foundation that the center's cyber-physical systems stand on.



Arash Rezaee: AI services over a software-defined, multi-layer network.



Ryan McCann: FUSION, reinforcement learning over a software-defined optical mesh.

The papers

- QoT-Aware Spectral and Spatial Scaling Trade-offs in Multi-Band Space Division Multiplexing over EONs
JOCN, Aug. 2026; invited extension of ECOC 2025. doi:10.1364/JOCN.596854
- QoT-Aware Dynamic Resource Allocation and Grooming in Multi-Band Space-Division Multiplexing Networks
JOCN, Oct. 2026; invited extension of ONDM 2025. doi:10.1364/JOCN.596435
- FUSION: A Unified Benchmarking Framework for Reproducible Optical Network Research
JOCN, Sept. 2026, feature issue on benchmarking in optical networks. doi:10.1364/JOCN.593123

UMass Lowell's contributions were supported in part by the National Science Foundation under Award No. 2008530.

August 2026 at the center

New awards

Planning Federated AI-Ready Cyberinfrastructure for Advanced Microscopy and Imaging: A Teach-Explore-Design Framework for Community-Driven Infrastructure \$499K

National Science Foundation, OAC Category III (Award #2609490). PI Lewis Tseng; Co-PIs Hsien-Yuan Hsu (UMass Lowell) and Yu-Tsun Shao (University of Southern California)

New papers

A. Moeinaddini, T. Zhang, C. D'Agostino, Y. Xie, Y. Zou.

Accounting for under-reporting in wildlife–vehicle collision hotspot identification using copulas and Bayesian model averaging. *Accident Analysis & Prevention*, vol. 233, art. 108583, Aug. 2026.

doi:10.1016/j.aap.2026.108583

A. Rezaee, F. Arpanaei, R. McCann, L. Nadal, J. A.

Hernández, V. M. Vokkarane. **QoT-Aware Spectral and Spatial Scaling Trade-offs in Multi-Band Space Division Multiplexing over EONs [Invited].** *IEEE/Optica Journal of Optical Communications and Networking*, vol. 18, no. 8, pp. C160-C172, Aug. 2026.

doi:10.1364/JOCN.596854

S. Xu, Y. Chen, Y. Xie, C. Wang. **Quantifying the safety effects of left-turn signal control mode: A heterogeneous causal inference framework.** *Accident Analysis & Prevention*, vol. 233, art. 108549, Aug. 2026.

doi:10.1016/j.aap.2026.108549

L. Tseng. **Timely Control for Quantum Cloud: Orchestrating Entanglement Under Decay.** *3rd ACM SIGCOMM Workshop on Quantum Networks and Distributed Quantum Computing (QuNet)*, ACM SIGCOMM 2026, pp. 32-34, Aug. 2026.

doi:10.1145/3833409.3833443

Milestones

Lewis Tseng presents timely control for quantum clouds at the ACM SIGCOMM 2026 QuNet workshop.

Coming up

- **September 30** New England Governors' Nuclear Energy Summit, hosted by UMass Lowell at the Rist Institute.
- **October 1** The NSF SUMMIT award begins; the search for its postdoctoral research associate is open.
- **2026 to 2027** SUMMIT opens to collaborators from utilities, vendors, agencies, and other universities.
- **2028** IEEE HPSR 2028 at South Asian University, New Delhi, with the director as TPC Co-Chair.

Work with us

Doctoral applicants: write to the faculty member whose work matches yours and copy the director. Companies and agencies: the laboratories page lists what each facility can offer, and the SUMMIT testbed opens to collaborators in 2026 to 2027.

Support the center

Gifts fund student travel, seed projects, and the capstone program. Give at smartcyberphysical.org, or write to the director.

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